

Soaiba Nuzhat

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EDUCATION

Columbia University, Fu Foundation School of Engineering and Applied Sciences

Graduation Date: May 2027

Bachelor of Science in Computer Science (Minor: Applied Mathematics)

GPA: 3.7/4.0

- **Relevant Coursework:** Agentic AI and Data Science (Graduate, A), Probabilistic Models and Machine Learning (Graduate), High Performance Machine Learning, AI (A), Data Structures and Algorithms, Operating Systems, Computer Architecture (A), Advanced Programming in C (A), Linear Algebra, Probability
- **Honors & Awards:** Dean's List, Kappa Alpha Theta National Scholarship, Summer at SEAS Scholarship

Oxford University, St. Anne's College

Spring 2026

Computer Science and Mathematics Program

Study Abroad

- **Relevant Coursework:** Machine Learning, Geometric Deep Learning, Compilers, Numerical Analysis

SKILLS

Languages & Tools: Python, Java, C, C++, SQL, Bash, MATLAB, React, Streamlit, Git, Linux, Neo4j, AWS

AI/ML: RAG, PyTorch, Agentic Architecture, MCP servers, Deep Learning (CNN/GNN/RNN/Attention), Neural Network Design, Fine-tuning, Model Training/Evaluation/Deployment, Benchmarking, Data Flow Control, NumPy, Pandas

EXPERIENCE

Pfizer

Data Engineering Intern

Clinical Data Science AI/ML Innovation Team

June 2026 – August 2026

- Boosted clinical-data query accuracy to 70%, a 10-point gain over a LangChain SQL-agent baseline (60%), by architecting a graph-based RAG agent over a 1.3M-node, 7M-relationship Neo4j knowledge graph fusing metadata and trial data across 5 clinical domains
- Secured a full-time team to continue the project, green-lit by the head of data science post-presentation, by benchmarking the graph agent across two evaluation types (a 4-domain data report and free-text queries) to prove it trades answer-agreement consistency (40% vs. 60%) for more thorough clinical review

Columbia University DAP Lab

AI Safety Researcher

Data Flow Control (DFC) team, advised by Charlie Summers and Dr. Eugene Wu

June 2026 – Present

- Proved coding agents can generate secure Python without security prompting, measured by a 20/20 pass rate on both functional and security tests, by benchmarking Claude Opus 4.8 across four adversarial prompt regimes (generate, generate-with-hints, fix, fix-with-hints) on Alibaba's multi-language coding-security benchmark
- Exposed latency as the hidden cost of security prompting, measured by a 3.5x jump in code-generation time (7.0s to 24.9s) with no gain in the 100% pass rate, by isolating prompt-variant effects to justify the team's shift toward deterministic data-flow controls over probabilistic prompting

Columbia University Medical Center

AI Researcher

Vibrational Imaging of Brain Elastography (VIBES) Lab, Dr. Grace McIlvain

June 2024 – December 2025

- Optimized synthetic brain-scan generation to under 5 minutes per 1,000+ scans by engineering a scalable virtual MRE simulator on a Columbia HPC cluster and MATLAB App Designer with real-time tuning across 10+ parameters
- Modeled cerebrospinal fluid (CSF) effects on brain elasticity, quantified by pre- and post-CSF stiffness maps computed from 1024×1024 imaging matrices, by developing computational MATLAB algorithms to process brain-imaging data

PROJECTS

Agentic AI Stock Recommender

August 2025 – Present

- Engineered a multi-agent LLM system generating buy/sell/hold equity recommendations, scoring $\geq 4.5/5$ on plausibility, signal coverage, and internal consistency in a 16-sample human study, by orchestrating 3 specialist agents (fundamental, technical, news) and a synthesis manager in Python with the CrewAI framework
- Built a fact-pack retrieval pipeline integrating 4 financial data sources (Compustat, CRSP, IBES, Capital IQ) across the Dow Jones 30 over 16 years (2008–2024); co-authoring an AAI-workshop paper advised by Dr. Alfio Gliozzo (IBM Chief Science Catalyst for Data & AI Platforms)

LEADERSHIP & INVOLVEMENT

Engineering Recruitment Committee, Co-Chair & Head Tour Guide

Sep 2023 – Present

Led a 75-guide program reaching 500+ prospective students, by onboarding 15 new guides, running annual all-crew training, and hosting weekly tours and Q&A panels alongside admissions officers.

Multivariable Calculus, Teaching Assistant (1 of 5 selected)

Sep 2023 – Dec 2023

Raised section exam performance 20% across 100+ students through weekly recitations and office hours.